

AI-Powered Personal Productivity Dashboard

Final Report

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Abstract

This report presents the design and development of a full-stack AI-powered productivity dashboard. The system enables users to manage tasks, reminders, and meetings while interacting with a locally hosted AI assistant powered by Ollama. It includes CRUD functionality, authentication, and a modern UI with responsive design.

Introduction

The project aims to build a personal productivity dashboard integrated with an AI assistant. It allows users to manage day-to-day productivity tasks and query an AI model for contextual insights.

Objectives

- Provide CRUD functionality for tasks, reminders, and meetings.
- Implement authentication (JWT/Custom).
- Integrate AI assistant (Ollama).
- Support dark mode and responsive UI.

- Enable deployment via Vercel (frontend) and Ngrok/VPS (backend).

System Architecture

The system consists of a React-based frontend, a Node.js/Express backend, and integration with Ollama AI running locally. Frontend communicates with the backend via REST APIs, which in turn connect to Ollama.

Technology Stack

Layer	Technology
Frontend	React + Vite + Tailwind CSS
Backend	Node.js + Express
AI Model	Ollama (LLaMA3, Mistral, Gemma)
Auth	JWT / OAuth / Custom
Deployment	Frontend: Vercel Backend: Ngrok / VPS
Data Storage	JSON / In-memory / DB (future)

System Modules

- User Interface (React + Tailwind): Sidebar navigation, task/reminder/meeting panels, chat modal, dark mode.
- Backend API (Node.js): Provides REST endpoints for authentication, tasks, reminders, meetings, and chat.
- AI Assistant (Ollama): Responds to user queries and summarizes tasks/reminders.
- Authentication: Login/Signup with JWT tokens.
- Deployment: Frontend on Vercel, backend on Ngrok/VPS.

AI Integration

The AI assistant is powered by Ollama models (e.g., LLaMA3). Prompts include user-specific context such as tasks and reminders, enabling personalized responses. The assistant can answer queries like 'What 's due today?' or 'List my meetings for tomorrow.'

Features

- CRUD for tasks, reminders, meetings.
- AI chat assistant with contextual awareness.
- Modern UI with dark mode.
- Mobile responsiveness.
- Protected routes with authentication.

Future Enhancements

- Stream AI responses in real-time.
- Store chat history in DB.
- Enable voice input.
- Dynamic updates with new user data.
- Secure login via OAuth.

Conclusion

The AI-powered productivity dashboard combines task management with AI assistance, providing a practical tool for boosting personal productivity. Its modular architecture ensures scalability for future enhancements.

Appendix: Folder Structure

/client (React) /components /pages /assets App.jsx index.jsx /server (Node.js) server.js routes/ chat.js tasks.js
auth.js data/ userData.json